PROJECT AND TEAM INFORMATION

Project Title

File Encryption & Decryption Tool

Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

Personal documents, educational records, source code, passwords, and a number of other similar kinds of information can be found in digital files. If these files are stored, sent or received without being protected, a stranger will be able to see the information easily. There is also the matter of file size since the bigger the files are, the more storage space and bandwidth they will take, both of which could become problems. The purpose of this project was to create an application, specifically a File Encryption & Compression Tool, in order to overcome the aforementioned difficulties. The application would enable the user to select a file, compress the data in order to save space, and then encrypt it in order to stop unauthorised access. When permission is granted, the application could decrypt and decompress the files so that access to the data they contain could be provided. Other benefits of carrying out such a project would be gaining a better understanding of file operations and data structures, of the use of bitwise operators, algorithms, and finally object oriented programming.

State of the Art / Current solution (1 pt)

At the present time, there are usually two different kinds of software—one for encrypting files and another for compressing them. Encryption permits files to be stored in a form that can only be decoded when the correct key or password is entered, whereas compression enables files to be stored in a more efficient format so that the final file becomes smaller. Since these functions are generally provided separately, it is often necessary to use more than one programme. Some of the programs that are widely available include a great many complicated and unnecessary features, which can confuse those who are new to the algorithms in question.

Project Goals and Milestones (2 pts)

Goals

- *Develop a tool to compress and encrypt files securely.*
- *Implement file input and output operations.*
- *Add compression, decompression, encryption, and decryption features.*

Milestones

- *Analyze requirements and design the system.*
- *Develop and test the compression module.*
- *Implement and test encryption and decryption.*

Project Approach (3 pts)

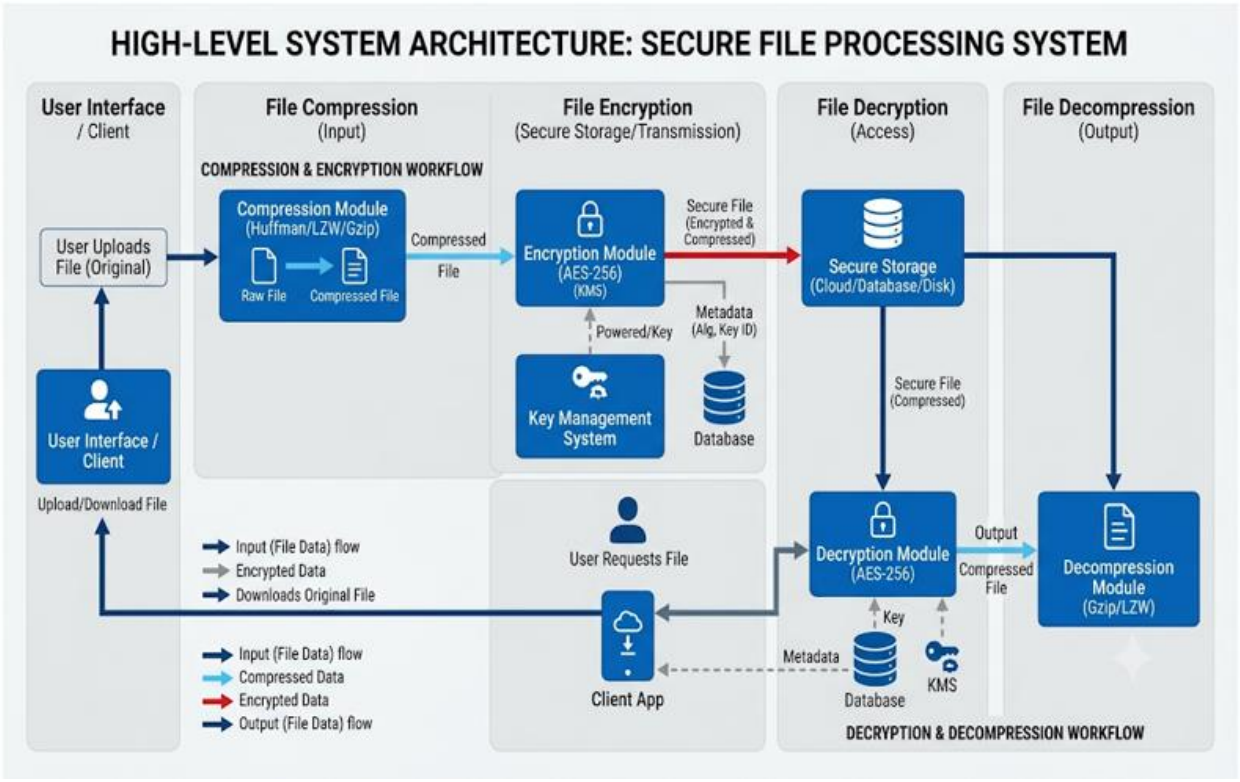
The application is developed using C and C++ and combines file handling, data structures, algorithms, bitwise operations, and object-oriented programming (OOP) to protect files securely.
The process of protecting a file begins with selecting the input file. The application then handles the file, compresses its data to reduce its size, and encrypts it before saving it as a protected file.

File Protection: *Input File → File Handling → Compression → Encryption → Protected File*
To recover the original file, the application first decrypts the protected file and then decompresses it to restore the original data.

File Recovery: *Protected File → File Handling → Decryption → Decompression → Original File*

The application is divided into different modules for file handling, compression, encryption and decryption, and user interface management. C is mainly used for file handling and implementing basic data structures, while C++ is used to organize the application through object-oriented programming.
We tested the application using different types of files to check whether encryption and decryption work correctly. We also verified that compression and decompression preserve the original file data. In addition, we tested how the application handles errors to ensure that files can be processed reliably and recovered without data loss.

System Architecture (High Level Diagram)



Project Outcome / Deliverables (1 pts)

The main aim of this project is to develop a user-friendly File Encryption and Compression Tool using C/C++. The application will allow users to select files and easily perform operations such as encryption, decryption, compression, and decompression.

The project will include the following features:

- *File Handling: Allows users to select files and manage file input and output operations.*
- *Compression Module: Reduces file size to save storage space.*
- *Encryption and Decryption Module: Protects files by encrypting them and allows users to restore them to their original form through decryption.*
- *Decompression and File Recovery: Restores compressed files and helps users retrieve their original data.*

Assumptions

The program uses input files that are accessible and readable by the user. It is designed for educational purposes to demonstrate how encryption and compression work, rather than to replace professional security or compression tools. A C/C++ compiler and the required development environment are also available. The original files can be recovered by the authorized owner using the correct encryption key or password.